MODULE 04

Statistical Thinking

DATA SCIENCE · LEARNING NOTES

Mohan Babu University · Course-aligned study guide

Concepts • examples • practice • revision

01. Representing histograms

Learning focus. This section introduces representing histograms as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to representing histograms. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

02. Outliers

Learning focus. This section introduces outliers as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to outliers. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

03. Summarizing distributions

Learning focus. This section introduces summarizing distributions as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to summarizing distributions. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

04. Variance and standard deviation

Learning focus. This section introduces variance and standard deviation as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to variance and standard deviation. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

05. Reporting results

Learning focus. This section introduces reporting results as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to reporting results. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

06. Probability mass functions

Learning focus. This section introduces probability mass functions as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to probability mass functions. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

07. Class size paradox

Learning focus. This section introduces class size paradox as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to class size paradox. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

08. DataFrame indexing

Learning focus. This section introduces dataframe indexing as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to dataframe indexing. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

09. Cumulative distribution functions

Learning focus. This section introduces cumulative distribution functions as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to cumulative distribution functions. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

10. Percentiles and ranks

Learning focus. This section introduces percentiles and ranks as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to percentiles and ranks. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

11. Random numbers

Learning focus. This section introduces random numbers as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to random numbers. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

12. Exponential distribution

Learning focus. This section introduces exponential distribution as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to exponential distribution. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

13. Normal and lognormal distributions

Learning focus. This section introduces normal and lognormal distributions as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to normal and lognormal distributions. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

14. Module review

Learning focus. This section introduces module review as part of statistical thinking. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to module review. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?